Convex Analysis

Exercise 1

Consider a transmitter at a location $\boldsymbol{x}_t = (-2, -4)$. It is required to find the optimal location to place a receiver $\boldsymbol{x}_r$ within a specified area of land such that the received signal's power is maximized. The received signal's power is inversely proportional to the squared distance between the transmitter and the receiver:

$$P(\boldsymbol{x}) \propto \frac{1}{d^2} \tag{1}$$

The allowable area to place the receiver is given by the following constraints in the euclidean plane:

$$c_1(\boldsymbol{x}) = -x_1^2 - (x_2 + 4)^2 + 16 \geq 0 \tag{2}$$
$$c_2(\boldsymbol{x}) = x_1 - x_2 - 6 \geq 0 \tag{3}$$

(a) Is the feasible region convex?

(b) Consider now an additional constraint defined as:

$$c_3(\boldsymbol{x}) = x_1^2 + (x_2 + 6)^2 \geq 2. \tag{4}$$

Is the feasible region convex?

Exercise 2

Consider the scalar function $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(\mathbf{x}) = \mathbf{x}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}\mathbf{A}\mathbf{x} + \mathbf{b}^{\mathrm{T}}\mathbf{x} \tag{5}$$

where

$$\mathbf{A} = \begin{bmatrix} \frac{1}{5} & -\frac{1}{3} \\ \frac{1}{20} & \frac{1}{5} \end{bmatrix} \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} \tag{6}$$

(a) Find the gradient function $g(\mathbf{x}) = \nabla f(\mathbf{x})$ of $f(\mathbf{x})$.

> *Hint*: $\frac{d}{d\mathbf{x}}(\mathbf{x}^{\mathrm{T}}\mathbf{Q}\mathbf{x} + \mathbf{b}^{\mathrm{T}}\mathbf{x} + \mathbf{c}) = \mathbf{Q}^{\mathrm{T}}\mathbf{x} + \mathbf{Q}\mathbf{x} + \mathbf{b}$

(b) Find the Hessian $H_f(\mathbf{x})$ of $f(\mathbf{x})$

(c) Is $f(\mathbf{x})$ convex? Why?

> *Hint*: What can you say about $\mathbf{A}^{\mathrm{T}}\mathbf{A}$?

(d) Find the Jacobian matrix for a linear function. That is, find $J_f(x)$ for $f(x) = Ax$ with $A \in \mathbb{R}^{n \times m}$.

(d) Repeat (a) and (b) from above but now for the function $f : \mathbb{R}^n \to \mathbb{R}$ given by

$$f(\mathbf{x}) = \|\mathbf{r}\|_2^2 = \mathbf{r}^\top \mathbf{r}$$

with $\mathbf{A} \in \mathbb{R}^{n \times m}$, $\mathbf{b} \in \mathbb{R}^n$, and $\mathbf{r} = \mathbf{b} - \mathbf{A}\mathbf{x}$ being the residuals between $\mathbf{A}\mathbf{x}$ and $\mathbf{b}$.

(e) What requirements on $\mathbf{A}$ would imply that $f(\mathbf{x})$ convex (and strictly convex)?

> *Hint*: Consider $\mathbf{A}^{\mathrm{T}}\mathbf{A}$?